CSA1701 - ARTIFICIAL INTELLIGENCE

6.Question: Write a Python program to solve the Vacuum Cleaner problem.

Aim: To simulate a vacuum cleaner that cleans two rooms when they are dirty.

Algorithm:

Initialize the condition of rooms A and B.

Place the vacuum cleaner in room A.

Check whether the current room is dirty.

Clean it if necessary.

Move to the next room and repeat.

Display the final room conditions.

Program:

6 python.py - C:/Users/praji/AppData/Local/Programs/Python/Python314/6 python.py (3.14.8)

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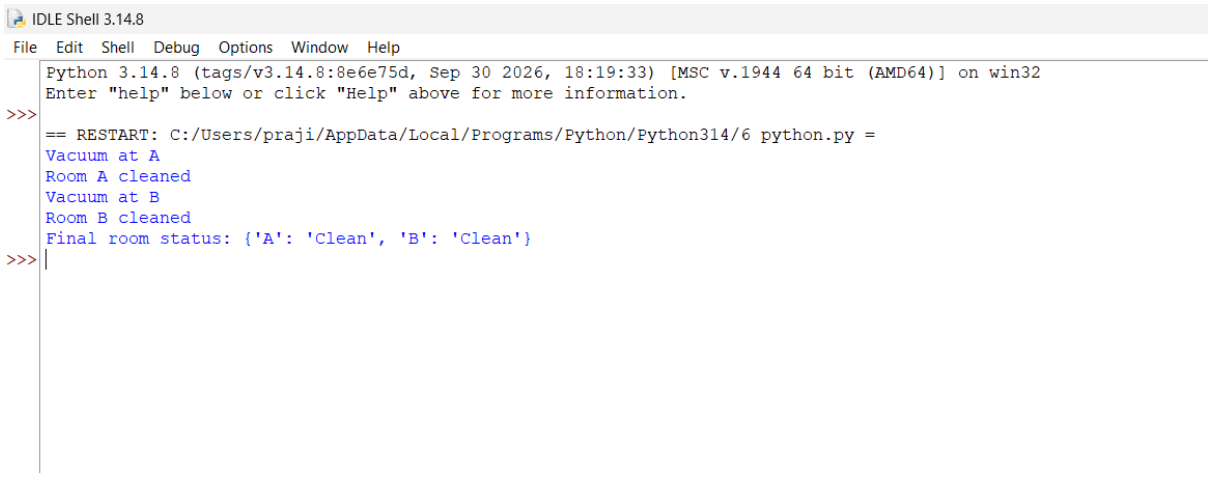
```
rooms = {"A": "Dirty", "B": "Dirty"}
location = "A"

while True:
    print("Vacuum at", location)

    if rooms[location] == "Dirty":
        rooms[location] = "Clean"
        print("Room", location, "cleaned")
    else:
        print("Room", location, "already clean")

    if location == "A":
        location = "B"
    else:
        break

print("Final room status:", rooms)
```



Result: Thus, the Vacuum Cleaner problem was simulated successfully, and both rooms were cleaned.